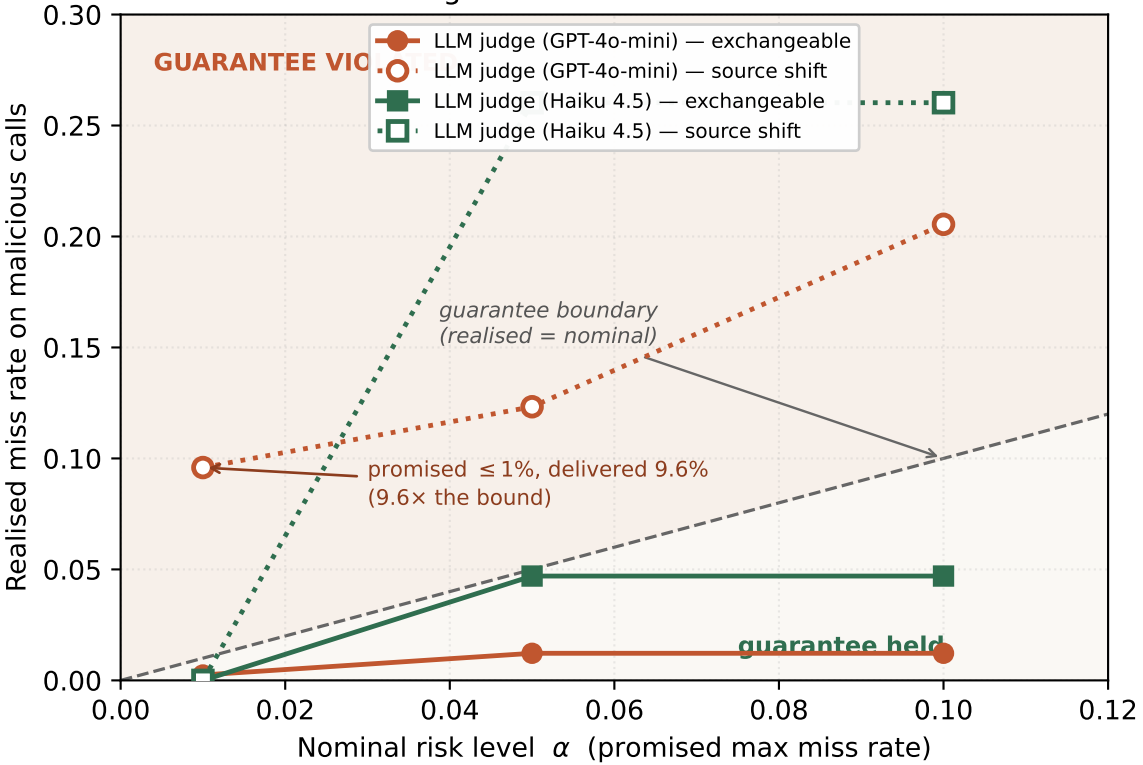
Conformal risk guarantees do not survive source shift
GUARANTEE VIO
LLM judge (GPT-4o-mini) — exchangeable
LLM judge (GPT-4o-mini) — source shift
LLM judge (Haiku 4.5) — exchangeable
LLM judge (Haiku 4.5) — source shift
guarantee boundary
(realised = nominal)
promised ≤ 1%, delivered 9.6%
(9.6× the bound)
guarantee held
Realised miss rate on malicious calls
Nominal risk level α (promised max miss rate)
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0.00
0.02
0.04
0.06
0.08
0.10
0.12